Temporal Relevance Engine

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HONDA

HRI
Honda Research Institute USA

LABS

Origin of 99P Labs' Name

- **Inspired by Soichiro Honda's quote:**

"Success represents the 1% of your work which results from the 99% that is called failure."

- **Philosophy:** Success is built on failure, not the opposite of it.

Research Thrusts

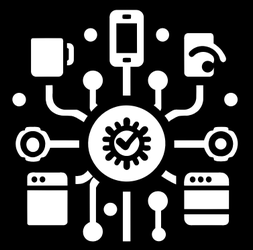
Joint Cognitive Systems: Enhancing cooperation and collaboration between humans and intelligent machines

Sustainable Energy & Materials: Advancing battery technologies and developing materials for better energy systems and sustainability

Computational Design and Digital Twins: Creating virtual models to address real-world challenges

Software-defined Intelligence: Building connected, intelligent platforms that adapt and evolve

Software-defined Intelligence



- **Embedded systems** provide the link between smart systems and the physical world, enabling real-world sensing and action

- **Connectivity & edge-to-cloud** manage the flow of intelligence across distributed systems, allowing flexible and scalable operations

- **Informative data** convert operational data into insights for continuous optimization and smarter decision-making

The Vision: The Big Problem We Hope to Solve

- **How might we help teams turn messy data into useful context for decisions and action?**

- “Useful context” means the right mix of:
  - what is true right now
  - what has been true over time
  - what was decided and why
- The key challenge is time. Some information changes fast. Other information stays useful for longer.
- If we handle time well, we expect it could help in places like:
  - ***Code repositories:*** support questions about current behavior while keeping key history.
  - ***Brand sentiment:*** understand what changed and whether it is lasting.
  - ***Innovation work:*** reuse past learning instead of repeating work.

- We are not trying to build all three solutions. These are examples of where this capability could matter.

The Two-Level Problem (Open for Debate)

Level 1: Why we care

- Build a better way to create context from data, so people can make better calls with less effort.
- The bet is that “time-aware context” is a missing piece across many kinds of data.

Level 2: What this project would build

- Build a “temporal relevance” layer for retrieval that picks the right mix of fresh and durable information for each question.
- Open for debate
- Your first job is to challenge this framing.
- If you think the real problem is different, say so and rewrite it.
- Make the problem your own, with clear boundaries and a clear definition of success.

What We're Asking From You

1) Clarify the problem
- Pressure-test the two-level framing.
- Define what "better context" means in plain terms.
- Pick one target setting to anchor the work (real or toy).

2) Define how we will evaluate
- A big part of the project is figuring out how to measure progress.
- Propose a test plan that can tell us if the context got more useful.
- Compare against simple baselines (like recency-only retrieval).

3) Build through experiments
- Run small tests and write down results.
- Turn the best idea into a working prototype.
- Document your journey so others can build on it.

What type of team member are we looking for?
- High agency is required.
- You will have wide freedom on approach.
- You own the plan, the pace, and the quality bar

Deliverables

1. An MVP (Minimum Viable Prototype) of the system.
2. The code (link to Github Repo).
3. A team blog post describing the project.

Relevant Links

Blog: medium.com/99p-labs

LinkedIn: linkedin.com/company/99p-labs

Website: www.99plabs.com

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